

Growing Fast, Failing Hotter

What is driving service failures, new-rider churn and margin erosion on VoltRelay's battery-swap network

+177%

completed swaps

4.4% → 8.3%

attempts not served

88% → 85%

new-rider retention

₹38 → ₹54

battery wear per swap

Prepared for

VoltRelay Energy — Operations & Customer Experience Leadership

Scope

Jan 2024 – Jun 2025 · 6 cities · 152 stations · 3.88 M swap attempts

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Executive summary

THE BOTTOM LINE

VoltRelay is growing fast, but three fixable problems are eating that growth. HOT: old chargers can't keep up in summer heat. WEAK: one battery brand wore out three times faster. CHEAP: the biggest partner pays too little. Fixing these will help more than opening new stations.

303 k

swaps per month

**▲ 177% since
early 2024****₹19.7 M**

revenue per month

**▲ 200% since
early 2024****8.3%**visits with no
battery**▲ doubled from
4.4%****84.5%**

new riders who stay

▼ from 88.4%**+₹1.7**change in profit per
swap**≈ flat despite
price rise**

Last 3 months of data (Apr–Jun 2025) compared with the first 3 months (Jan–Mar 2024).

The three problems

HOT

Old chargers fail in summer heat

On days above 38°C, three times more riders find no charged battery (15% vs 4%). Old Gen1 chargers take up to 194 minutes per battery in hot cabinets instead of 88.

WEAK

One battery brand wore out 3× faster

Kyron 2W batteries (from Sep 2024) lost health three times faster and gave 48 km instead of 61 km. The extra battery cost wiped out the price rise.

CHEAP

Biggest partner, lowest profit

ZipDrop's discount went from 12% to 28% in Nov 2024. It is now the largest partner but the least profitable one.

New riders feel these problems first. Those who get worn batteries or are turned away several times in their first two weeks, and those who join in summer, are **6–11 points less likely to stay**.

What we recommend

#	Action	When
1	Upgrade old Gen1 chargers and add cooling at the 15–20 worst stations in hot cities	Before April
2	Replace Kyron batteries and test every new battery supplier before buying at scale	Now
3	Protect new riders' first 14 days: give them healthy batteries and reliable stations	Next 60 days
4	Renegotiate ZipDrop's discount (about 15%) and do not sign an exclusive deal	Next renewal
5	Use peak pricing to earn more , not to fix crowding; charge it to all partners	Ongoing

1

The business, key words & our approach

What does VoltRelay do, what did leadership ask, and how did we answer?

The business in one minute. VoltRelay runs battery-swap stations for delivery and bike-taxi riders in six Indian cities. A rider arrives with an empty battery, puts it in a cabinet and takes a charged one in about two minutes. The cabinet then recharges the empty battery for the next rider. Riders pay per swap, with a prepaid pack, or through their employer (a “fleet partner”) at a discount.

What leadership noticed. Swaps and revenue grew strongly from January 2024 to June 2025, but more riders were turned away, fewer new riders stayed, and profit per swap did not improve. Before deciding next year's budget, leadership asked six questions:

Leadership's question	Answered in
How has the network performed over time?	Section 2
When and where do riders get turned away, and how does that affect their experience?	Section 3
Do station type, location and age matter?	Section 4
How does battery health matter, and does any battery group stand out?	Section 5
How did prices, the peak-pricing pilot and partner contracts affect profit?	Section 6
Why do new riders stop using VoltRelay?	Section 7
Which of four budget proposals should be funded?	Section 8

Key words used in this report

Word	Meaning
Swap / visit	A rider coming to a station to exchange an empty battery for a charged one
Visit with no battery (not served)	A visit that ends without a swap: no charged battery, rider gave up waiting, cancelled, or system error
Battery health (SoH)	How much of its original capacity a battery still holds. 100% = new; lower = less range
Gen1 / Gen2 / Gen3	Charger generations at a station. Gen1 is the oldest, Gen3 the newest and fastest
2W / 3W	Two-wheelers (scooters, bikes) and three-wheelers (cargo autos)
Revenue	All the money VoltRelay receives from swaps
Profit per swap	Money from one swap minus its costs: electricity, battery wear and station rent (see Section 6)
New riders who stay (retention)	New riders who still make swaps 1-2 months after their first swap

How we did the analysis

1. Clean	2. Connect	3. Track	4. Dig in	5. Test	6. Recommend
Fixed 8 data problems	Linked 8 data tables	Monthly trends	By hour, city, station, battery	Before/after & statistical checks	Turn evidence into actions

Cleaning the data first

Real data has errors. The organizers hid several on purpose. We found and fixed each one before any analysis:

Problem found	What we did	How many
Internal test stations mixed into real data	Removed	62,031 records
Firmware bug saved times 5 h 30 min too early (Mar–Apr 2025)	Corrected the times	141,280 records
The same swap saved twice after a network drop	Removed the copy	1,370 records
Impossible readings (negative km, battery above 100%)	Removed or capped	21,140 readings
21 spellings for 6 cities (BLR, Bombay, Gurgaon...)	Standardised	All riders
Missing station sensor data; biased customer ratings	Kept blanks as blanks; didn't average ratings blindly	—

Detailed evidence for each fix is in Appendix A and in the notebook.

2

Performance over time

Question 1

THE QUESTION

How has the network performed over the 18 months? Do swaps, revenue, failures and profit all tell the same story?

SHORT ANSWER

No. Swaps and revenue tripled, but the share of visits where riders found no battery doubled, with big spikes every summer, and profit per swap stayed flat.

Every summer, many more riders are turned away

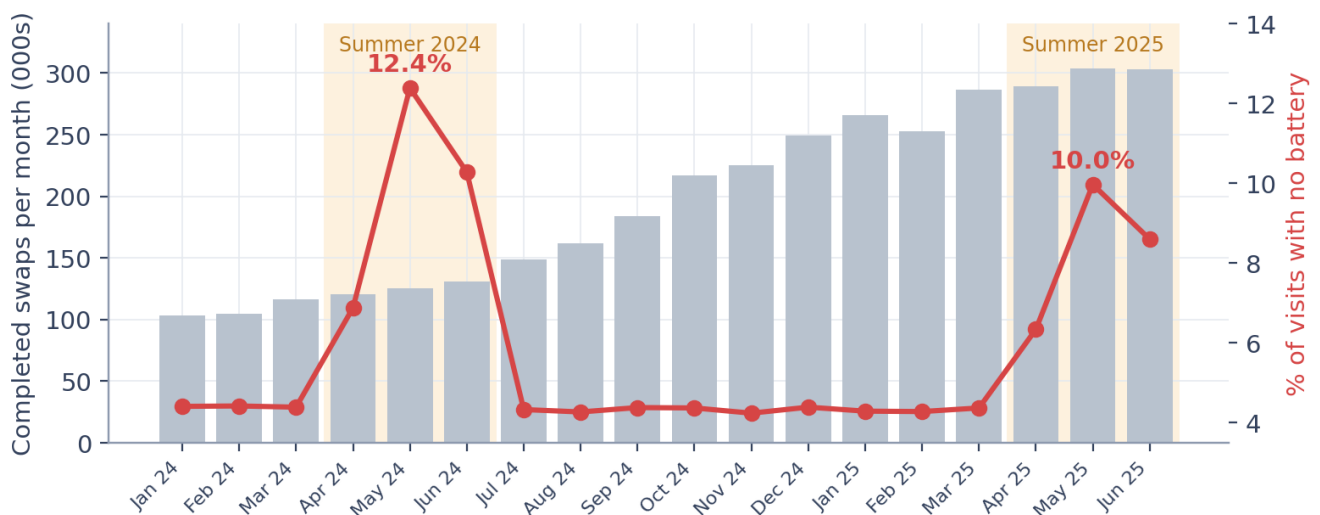


Figure 1. Grey bars: completed swaps per month. Red line: % of visits that ended with no battery. Shaded areas: summer months.

- Swaps grew from about 104,000 to 303,000 a month (+177%) and revenue by 200%, as stations grew from 90 to 150.
- Normally about 4 in every 100 visits end with no battery. In May 2024 it was 12 in 100, and in May 2025 it was 10 in 100.

- Prices rose in July 2024 (2W ₹60 → ₹65, 3W ₹100 → ₹110), but profit per swap did not stay higher. Section 6 explains why.

What this means for VoltRelay: the growth numbers hide a quality problem that comes back every summer.

3

Failures & customer experience

Question 2

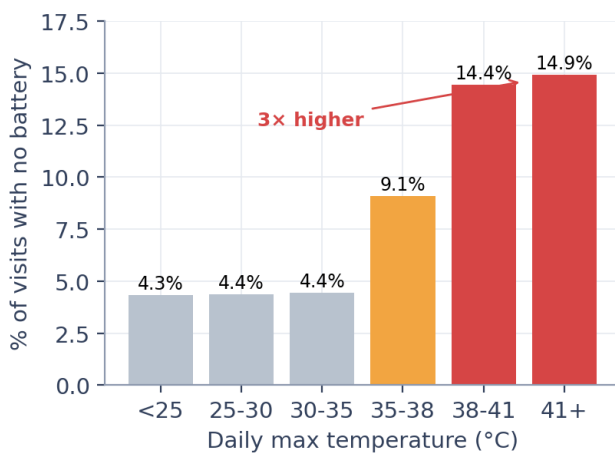
THE QUESTION

When and where do riders get turned away, and how does it affect their experience?

SHORT ANSWER

Mostly on hot days at stations with old Gen1 chargers. Heat makes these chargers so slow that batteries aren't ready in time. Complaints are highest in exactly the same hot cities.

Hot days triple failures...



...because old chargers slow down in heat

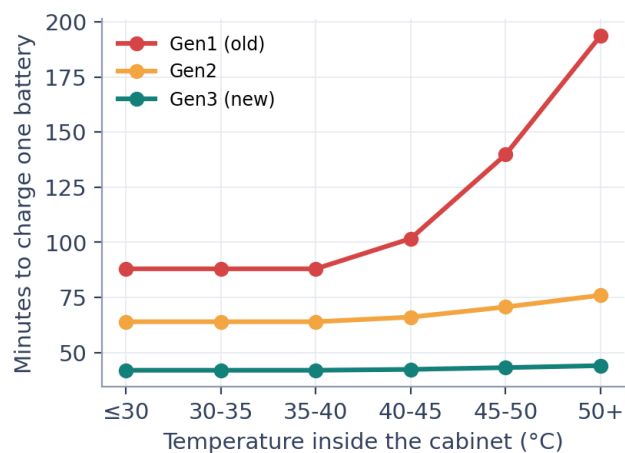


Figure 2. Left: % of visits with no battery by the day's highest temperature. Right: minutes needed to charge one battery, by temperature inside the cabinet and charger type.

- Below 35°C, about 4.4% of visits fail. Above 38°C, about 14.9% fail, three times more.
- Above 45°C inside the cabinet, an old Gen1 charger needs up to 194 minutes per battery (normally 88). New Gen3 chargers stay at about 43 minutes.
- The 15 worst stations are all old Gen1 stations in Jaipur, Delhi NCR and Hyderabad. The worst 10% of stations cause 24% of all failures.
- 3W riders are turned away twice as often (10.4% vs 5.5%), and 7-10 pm is the weakest time. Queue waits are normal (about 3.5 minutes), so the problem is no battery, not long queues.

What customers told us (support tickets)

Complaints follow failures: hot cities complain most

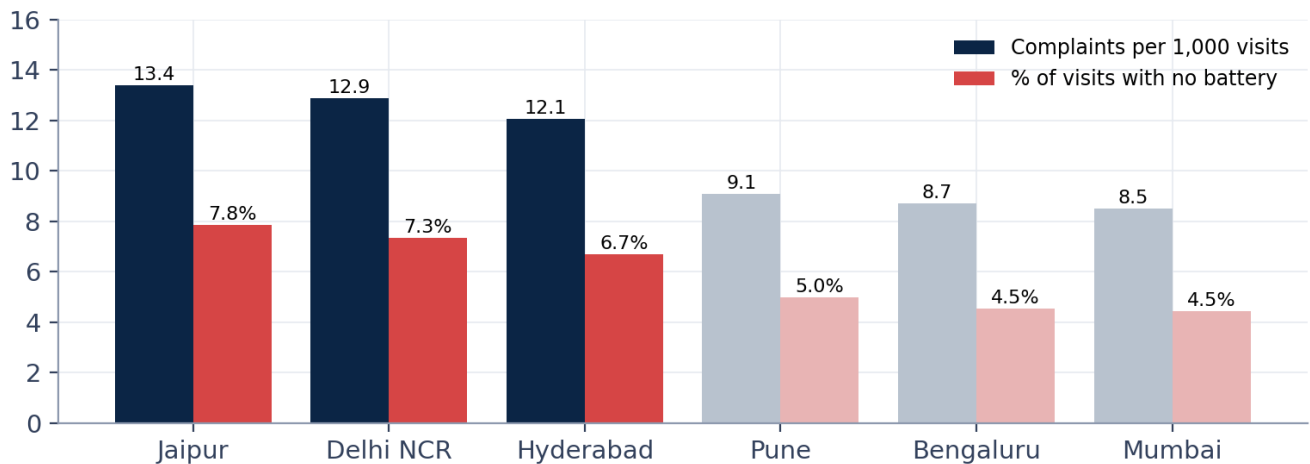


Figure 3. Dark bars: support tickets per 1,000 visits. Red bars: % of visits with no battery. Both are highest in the same three hot cities.

- Riders in Jaipur, Delhi NCR and Hyderabad raise 12–13 complaints per 1,000 visits, compared with 8.5–9 in Mumbai and Bengaluru.
- Complaint comments (in English and Hinglish) are mostly about batteries: 88% of “low range” and 70% of “no battery available” comments mention the battery or range.
- Ratings can mislead: only about 20% of slowly solved tickets get a rating, compared with 43% of quickly solved ones. So the average rating (about 3.6 out of 5) looks better than reality.
- Silent riders are the real risk: new riders who complained in their first month stayed more (90.1%) than those who didn't (85.9%). Many unhappy riders simply leave without complaining.

What this means for VoltRelay: it cannot wait for complaints. It should track failed visits directly, station by station, especially in summer.

4

Stations & cities

Question 3

THE QUESTION

Stations differ in charger type, location and opening date. Which of these differences matter?

SHORT ANSWER

Charger age and city climate matter; location type and how busy a station is do not. New stations were mostly opened in easy locations, while the failing old stations were not upgraded.

Hot cities and old chargers fail most

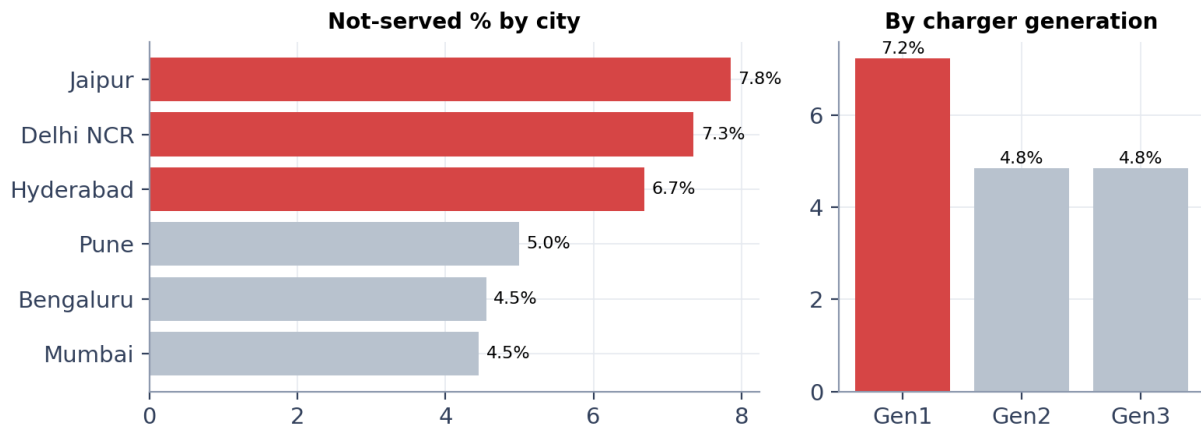


Figure 4. % of visits with no battery by city (left) and by charger type (right).

Station difference	What we found	Does it matter?
Charger type	Gen1 7.2% vs Gen2/Gen3 4.8%. Charging time: 92 / 64 / 42 minutes	Yes, strongly
City (climate)	Jaipur 7.9%, Delhi NCR 7.4%, Hyderabad 6.7% vs Mumbai 4.5%	Yes, strongly
Opening date (expansion wave)	New stations 4.9–5.0% vs original 6.2%, because they have newer chargers	Only through charger type
Location type, host, internet quality	Differences smaller than 1 point	No
How busy a station is	Busy and quiet stations fail at similar rates (5.5–6.1%)	No

- 27 of the 30 stations opened in late 2024 were at highway fuel pumps or residential sites: easy to open, but with only half the visits of original stations (about 17,000 vs 34,000 each).
- The old Gen1 stations that fail most were not upgraded, so the summer problem came back in 2025.

What this means for VoltRelay: upgrading old chargers will help riders more than opening new stations.

5

Batteries

Question 4

THE QUESTION

How does battery health affect riders, and does any battery group stand out?

SHORT ANSWER

Worn batteries give much less range, so riders have to swap more often. One brand, Kyron (2W), wore out three times faster than the others.

Worn batteries give fewer km, and Kyron wears out 3× faster

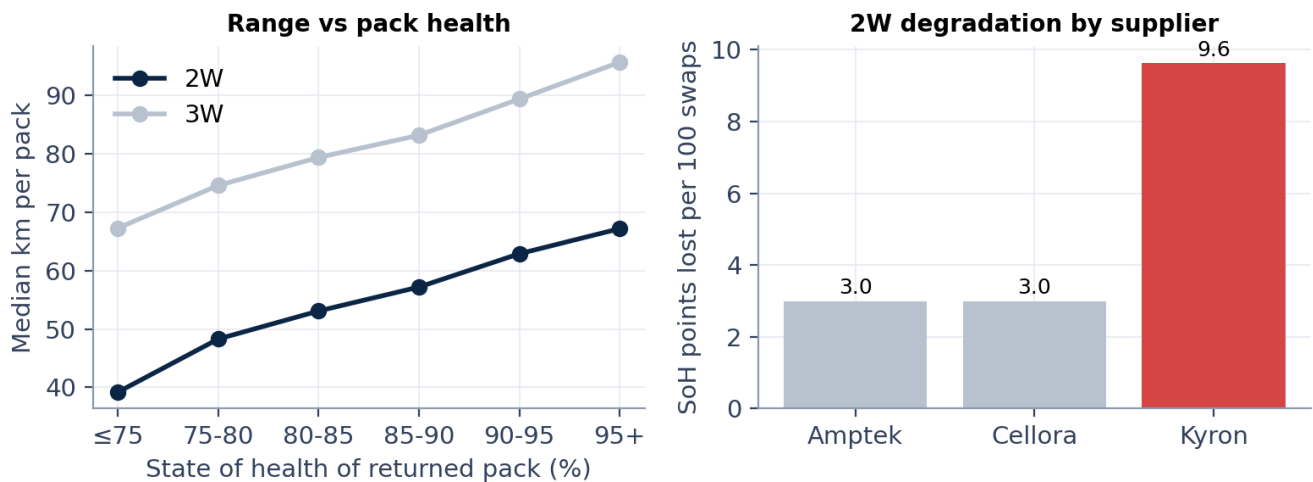


Figure 5. Left: km a rider gets from one battery, by battery health. Right: health lost per 100 swaps for 2W batteries, by brand.

- A healthy 2W battery (95%+ health) gives about 67 km. A worn one (below 75%) gives only about 39 km.
- Kyron 2W batteries lost 9.6 health points per 100 swaps, compared with 3.0 for Cellora and Amptek. All 1,385 are already retired, although the first only started in September 2024.
- Riders now get about 50 km per swap, compared with 72 km at the start, so they come back more often. That adds pressure on stations in summer.
- Three newer batches (AM-2506, AM-2505, CE-2505) only look fast-wearing because they are new. Per swap they are normal, so they are not a problem.

What this means for VoltRelay: battery quality, not the number of batteries, is the problem to fix.

6

Pricing, profit & partners

Question 5

THE QUESTION

Why didn't profit per swap improve, even though prices went up?

SHORT ANSWER

Because batteries started wearing out much faster. The price rise and cost savings added about ₹19 per swap, but faster battery wear and bigger discounts took away about ₹17. The result: profit per swap barely moved (+₹1.7).

REVENUE vs PROFIT: WHAT'S THE DIFFERENCE?

Revenue is all the money VoltRelay receives (₹19.7 M in June 2025). It is **not** profit. **Profit per swap** is what is left from one swap after paying its costs:

	Per swap, June 2025
Money in (after discounts)	₹65
– Battery wear (each swap uses up part of a battery's life)	₹54
– Electricity to recharge the battery	₹17
– Station rent & maintenance (shared across swaps)	₹12
= Profit per swap (estimate)	about –₹18

Battery wear is our estimate, so the exact profit level is uncertain. The **change over time** is reliable, and that is what the chart below shows.

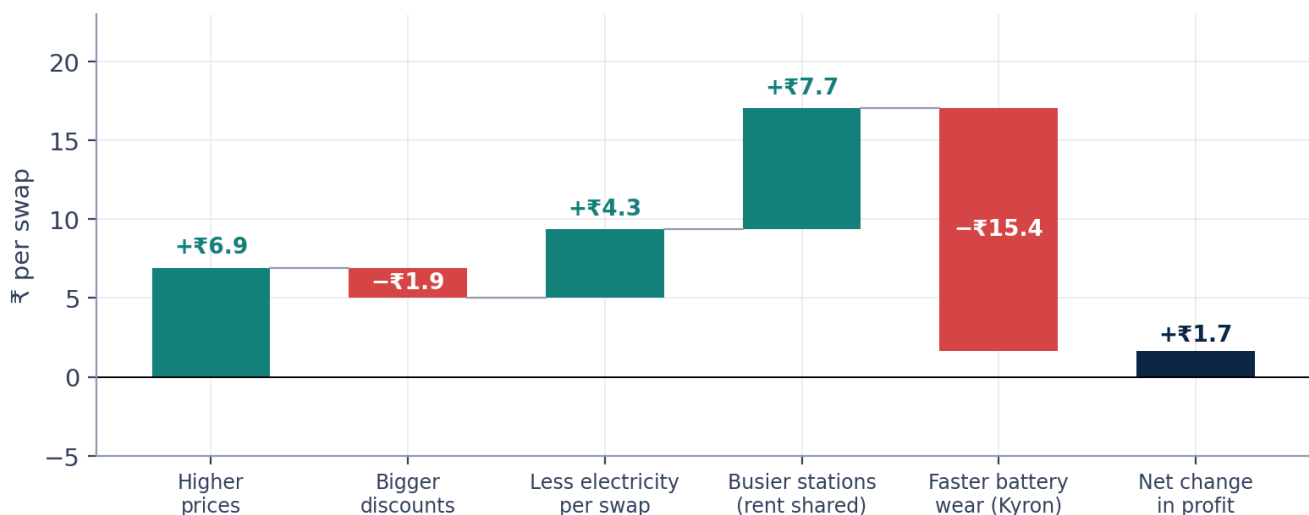
Where the price rise went (₹ per swap, Jan-Mar 2024 vs Apr-Jun 2025)

Figure 6. Green bars added profit per swap; red bars took it away. The dark bar is the net change.

- Gains: higher prices (+₹6.9), busier stations sharing their rent across more swaps (+₹7.7), and less electricity per swap (+₹4.3).
- Losses: faster battery wear (-₹15.4), which jumped in September 2024 when Kyron batteries arrived, and bigger discounts (-₹1.9), mostly ZipDrop's.
- Profit per swap peaked right after the price rise (July–August 2024) and then fell by about ₹10 once battery wear jumped.

The peak-hour pricing pilot

From 1 October 2024, Bengaluru and Pune charged more at busy times (12–2 pm and 7–10 pm) and less at quiet times. We compared these two cities with the other four, before and after the pilot:

What we measured	Change caused by the pilot	Verdict
Share of swaps at busy times	–2.6 points (riders barely moved)	Small effect
Riders turned away at busy times	+0.2 points (no improvement)	No effect
Money per swap	+₹5.1	Works

Fleet and independent riders reacted in the same small way. **Peak pricing earns more money, but it does not reduce crowding.**

Fleet partners: biggest is not best

The biggest partner earns VoltRelay the least per swap

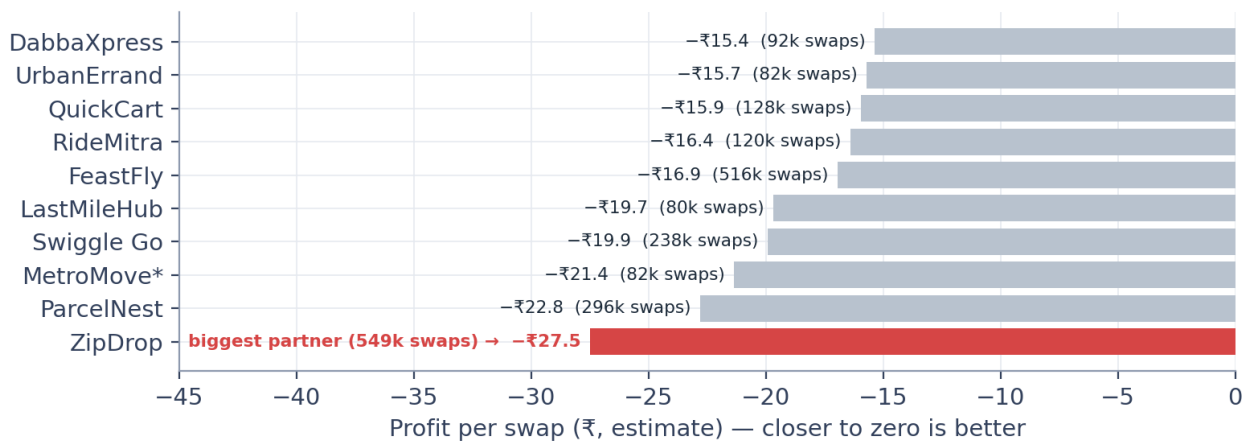


Figure 7. Estimated profit per swap for 2W fleet partners (closer to zero is better), with total swaps. *MetroMove has mixed 2W/3W riders. 3W cargo partners are lower (about –₹39) because 3W batteries cost more.

- ZipDrop has the most swaps (549,000) but the lowest profit per swap. After its discount rose from 12% to 28% in November 2024, money per swap fell from ₹57.5 to ₹48.6 while its swaps doubled. So every extra ZipDrop swap made the loss bigger.
- ZipDrop and Swiggle Go don't pay the peak-time surcharge, although about two-thirds of their swaps are at peak times.
- The most profitable partners per swap are DabbaXpress, UrbanErrand, QuickCart and RideMitra.

What this means for VoltRelay: fix battery quality first, renegotiate ZipDrop, and treat peak pricing as a way to earn more, not to fix crowding.

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Why new riders leave

Question 6

THE QUESTION

What makes new riders stop using VoltRelay? Which reasons are main and which are smaller?

SHORT ANSWER

A bad first two weeks: getting worn batteries or being turned away several times. This happens most to riders who join in summer.

We followed 17,393 new riders who joined between January 2024 and April 2025. Overall, 86.9% were still making swaps one to two months later, but this fell from 88.4% for the earliest riders to 84.5% for the latest.

New riders who have a bad start leave more

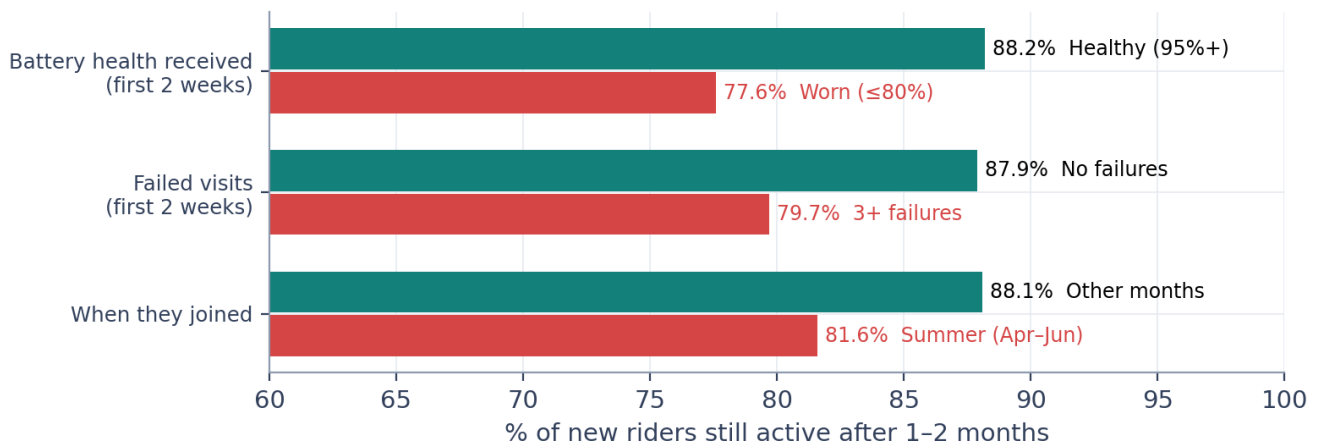


Figure 8. % of new riders still active 1–2 months after their first swap. Green: good first two weeks. Red: bad first two weeks.

Type	Reason	What we found
MAIN	Got worn batteries in the first 2 weeks	77.6% stayed vs 88.2% with healthy batteries (the strongest reason)
MAIN	Turned away 3+ times in the first 2 weeks	79.7% stayed vs 87.9% with no failures
MAIN	Joined in summer (when failures peak)	81.6% stayed vs 88.1% in other months
SMALLER	3W riders	Stay less, because they are turned away twice as often
SMALLER	Competitor station within 1.5 km	81.5% stayed vs 87.0% (a small group of 238 riders)
SMALLER	First station is an unreliable one	About 3–4 points lower
NOT A REASON	ID check (KYC), sign-up channel, fleet vs independent, city	Little or no difference once other reasons are considered

We checked all reasons together in one statistical model to separate main reasons from smaller ones (Appendix B).

What this means for VoltRelay: protect every new rider's first two weeks with healthy batteries and reliable stations, especially in summer.

8

What VoltRelay should do

Which of the four budget proposals does the evidence support?

THE QUESTION

Leadership proposed four uses for next year's budget: more stations, more batteries, peak pricing everywhere, and an exclusive deal with the largest partner. Which should it fund?

SHORT ANSWER

Fund reliability, not reach. Upgrade old chargers in hot cities and fix battery quality before adding stations, spreading peak pricing, or locking in the largest partner.

Proposal	What the data shows	Verdict	What to do instead / as well
More stations	How busy a station is doesn't predict failures. New stations went to easy, quiet sites. Failing old stations weren't upgraded.	REDIRECT	Upgrade Gen1 chargers to Gen3 and add cooling at the 15–20 worst hot-city stations before April
More batteries	The problem is quality, not quantity. Kyron batteries wore out 3× faster and gave about 20% less range.	FIX INSTEAD	Replace Kyron batteries, test new suppliers before buying at scale, and stock extra 3W batteries for summer
Peak pricing everywhere	+₹5.1 per swap, but riders barely moved and failures didn't improve.	LIMITED	Keep it as a way to earn more, charge it to all partners, but don't expect it to fix crowding
Exclusive deal with largest partner	ZipDrop earns VoltRelay the least per swap after its 28% discount.	DON'T FUND	Renegotiate the discount to about 15% and grow the more profitable partners
Extra: new riders	A bad first two weeks is the main reason new riders leave.	FUND	"First 14 days" program: healthy batteries and reliable stations for every new rider

Roadmap

Now (next 30 days)	Before summer (by March)	Next 6 months
- Stop using and replace Kyron batteries - Start the "first 14 days" battery rule - Open talks with ZipDrop	- Upgrade chargers + add cooling at the worst hot-city stations - Stock extra batteries in Jaipur, Delhi NCR, Hyderabad - Prepare a heatwave action plan	- Test every new battery supplier - Charge peak surcharge to all partners - Track monthly: % visits with no battery, battery health, new riders who stay, profit per swap

Expected impact

- Removing the summer spike alone would cut riders turned away in a year by roughly a quarter, and protect summer joiners, who currently stay 6.5 points less.
- Removing the extra Kyron battery wear would recover about ₹15 per swap, roughly ₹4.5 M a month at today's volume.
- Healthy batteries and reliable stations in the first two weeks tackle both main reasons new riders leave.

Appendix

A. Data cleaning: how each problem was detected

Issue	How we detected it	Treatment	Impact
Test stations (STN-TST*)	ID prefix; zero-value transactions at unusual hours	Excluded	62,031 events
Firmware v3.2.0 timezone bug	Hour-of-day pattern at affected stations shifted by 5.5 h; after correction, lunch and dinner peaks match other stations	+5 h 30 m for v3.2.0 events, 10 Mar – 14 Apr 2025	141,280 events
Offline-sync near-duplicates	Same rider + station + event type within 3 min, second record via offline_batch. It cannot be a real second swap (the battery returned is not the one just issued)	Later record dropped	1,370 events
Odometer resets	km_since_last_swap negative or above the 99.9th percentile (371 km)	Set to missing	11,456 readings
Sensor calibration drift	Battery health / charge readings above 100%	Capped at 100%	9,684 readings
City spellings	21 variants for 6 cities	Mapped to 6 cities	100% mapped
Telemetry gaps	status partial/missing: 10.9% at poor-connectivity sites vs 1.0% at good ones	Kept missing, never zero	'ok' hours only
Ratings (CSAT) not missing at random	Rating response rate 43% for tickets solved in under 24 h vs about 21% for over 1 day	Not averaged blindly	Reported by resolution time

B. Methods

- **Profit per swap.** Electricity = kWh to recharge × station electricity tariff. Station cost = monthly rent + maintenance ÷ that station's swaps that month. Battery wear = battery price × (health lost per swap ÷ usable health down to an assumed 70% end-of-service). The exact level depends on the 70% assumption; the trends and their timing do not.
- **Why failures happen.** Swap attempts were grouped by station and hour and matched to hourly station sensor data (cabinet temperature, charged batteries in stock, charging time), using only hours with complete sensor data.
- **Peak-pricing pilot.** Difference-in-differences: Bengaluru and Pune vs the other four cities, 60 days before vs 60 days after 1 Oct 2024.
- **New-rider retention.** Retention = at least one swap 31–60 days after the first swap. Each factor was first compared on its own, then all together in a logistic regression (controlling for city, sign-up channel and vehicle type). Main reasons had strong, consistent effects: battery health received ($z \approx 6.8$), summer start ($z \approx -6.1$). 3W riders: odds of staying ×0.66.

C. Limitations

This is observational data, so it shows strong links, not proof of cause; we call something a main driver only when several views of the data agree. Battery wear relies on an assumed end-of-service battery health, and complaint comments were screened with keywords only. The dataset is synthetic; company names are fictional.

AI DISCLOSURE

Analysis code, charts and this report were prepared with the help of Claude (Anthropic) and reviewed by the team. Fully reproducible code: VoltRelay_Final_Analysis.ipynb.